Mathematics in Java

Basic Mathematical Operations

Arithmetic Operators

Java provides several arithmetic operators for performing mathematical calculations:

- `+` Addition
- `-` Subtraction
- `*` Multiplication
- `/` Division
- `%` Modulus (remainder)

```java
int a = 10;
int b = 3;

int sum = a + b;        // 13
int difference = a - b; // 7
int product = a * b;    // 30
int quotient = a / b;   // 3
int remainder = a % b;  // 1
```

Order of Operations

Java follows the standard mathematical order of operations (PEMDAS):

1. Parentheses
2. Exponents (using Math.pow())
3. Multiplication and Division (left to right)
4. Addition and Subtraction (left to right)

```java
int result = 2 + 3 * 4;     // 14 (not 20)
int result2 = (2 + 3) * 4;  // 20
```

Working with Different Number Types

Integer Division vs. Decimal Division

```java
// Integer division
int x = 7;
int y = 2;
int intResult = x / y;  // 3 (truncated)

// Decimal division
double a = 7.0;
double b = 2.0;
double doubleResult = a / b;  // 3.5

// Mixed division
double mixedResult = (double) x / y;  // 3.5 (casting)
```

Data Type Considerations

```java
int num1 = 5;
int num2 = 2;
double result1 = num1 / num2;           // 2.0 (integer division first)
double result2 = (double) num1 / num2;  // 2.5 (proper decimal division)
```

Math Class Methods

Common Mathematical Functions

```java
// Square root
double sqrt = Math.sqrt(25);  // 5.0

// Power
double power = Math.pow(2, 3);  // 8.0

// Absolute value
int abs = Math.abs(-10);  // 10

// Maximum and minimum
int max = Math.max(5, 10);  // 10
int min = Math.min(5, 10);  // 5

// Rounding
double round = Math.round(3.7);  // 4.0
double ceil = Math.ceil(3.1);    // 4.0
double floor = Math.floor(3.9);  // 3.0
```

Trigonometric Functions

```java
// Convert degrees to radians
double degrees = 90;
double radians = Math.toRadians(degrees);

// Trigonometric functions
double sin = Math.sin(radians);  // 1.0
double cos = Math.cos(radians);  // 0.0 (approximately)
double tan = Math.tan(Math.toRadians(45));  // 1.0
```

Random Number Generation

Using Math.random()

```java
// Random decimal between 0.0 and 1.0
double random = Math.random();

// Random integer between 1 and 10
int randomInt = (int) (Math.random() * 10) + 1;

// Random integer between min and max (inclusive)
int min = 5;
int max = 15;
int randomRange = (int) (Math.random() * (max - min + 1)) + min;
```

Using Random Class

```java
import java.util.Random;

Random rand = new Random();
int randomNumber = rand.nextInt(100);  // 0 to 99
int randomRange = rand.nextInt(50) + 1;  // 1 to 50
```

Mathematical Formulas and Calculations

Area Calculations

```java
// Circle area
double radius = 5.0;
double circleArea = Math.PI * Math.pow(radius, 2);

// Rectangle area
double length = 10.0;
double width = 6.0;
double rectangleArea = length * width;

// Triangle area
double base = 8.0;
double height = 4.0;
double triangleArea = 0.5 * base * height;
```

Distance Formula

```java
// Distance between two points (x1, y1) and (x2, y2)
double x1 = 1.0, y1 = 2.0;
double x2 = 4.0, y2 = 6.0;
double distance = Math.sqrt(Math.pow(x2 - x1, 2) + Math.pow(y2 - y1, 2));
```

Quadratic Formula

```java
// For equation ax² + bx + c = 0
double a = 1.0, b = -5.0, c = 6.0;
double discriminant = Math.pow(b, 2) - 4 * a * c;

if (discriminant >= 0) {
    double root1 = (-b + Math.sqrt(discriminant)) / (2 * a);
    double root2 = (-b - Math.sqrt(discriminant)) / (2 * a);
    System.out.println("Root 1: " + root1 + ", Root 2: " + root2);
} else {
    System.out.println("No real roots");
}
```

Number Formatting and Precision

Controlling Decimal Places

```java
import java.text.DecimalFormat;

double number = 3.14159;
DecimalFormat df = new DecimalFormat("#.##");
String formatted = df.format(number);  // "3.14"

// Using printf
System.out.printf("%.2f%n", number);  // 3.14
```

Scientific Notation

```java
double largeNumber = 1234567.89;
System.out.printf("%.2e%n", largeNumber);  // 1.23e+06
```

Quiz Questions

1. What is the result of the expression `7 / 2` in Java when both operands are integers? a) 3.5 b) 3 c) 4 d) 3.0

Answer: b) 3

2. Which Math class method would you use to find the square root of a number? a) Math.power() b) Math.sqrt() c) Math.square() d) Math.root()

Answer: b) Math.sqrt()

3. What does the expression `Math.random()` return? a) A random integer between 0 and 10 b) A random decimal between 0.0 and 1.0 (excluding 1.0) c) A random integer between 1 and 100 d) A random decimal between 1.0 and 10.0

Answer: b) A random decimal between 0.0 and 1.0 (excluding 1.0)

4. What is the result of `10 % 3` in Java? a) 3 b) 3.33 c) 1 d) 0

Answer: c) 1

5. Which of the following correctly calculates 2 raised to the power of 3? a) Math.power(2, 3) b) Math.pow(2, 3) c) Math.exp(2, 3) d) 2^3

Answer: b) Math.pow(2, 3)

Practice Coding Questions

1. Write a Java program that calculates the area and perimeter of a rectangle. The program should prompt the user to enter the length and width, then display both the area and perimeter.

2. Create a program that generates a random number between 1 and 100, then asks the user to guess the number. The program should tell the user if their guess is too high, too low, or correct.

3. Write a program that calculates the hypotenuse of a right triangle using the Pythagorean theorem ($c^2 = a^2 + b^2$). The program should accept the lengths of the two sides and output the hypotenuse length.

4. Create a program that converts temperature from Celsius to Fahrenheit and vice versa. The formula for Celsius to Fahrenheit is: $F = (C \times 9/5) + 32$. The formula for Fahrenheit to Celsius is: $C = (F - 32) \times 5/9$.

5. Write a program that calculates compound interest using the formula: $A = P(1 + r/n)^{(nt)}$, where P is the principal amount, r is the annual interest rate, n is the number of times interest is compounded per year, and t is the number of years.